OUMALKAIRE ABOUBAKAR ABDI

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PROFILE

Computer Science graduate (GPA 3.7/4.0) who builds reproducible machine-learning and data-analysis pipelines across audio signal processing, computer vision and geospatial biodiversity data. Experienced in Python, R and SQL, rigorous model evaluation, survey-based research and documenting methods so results can be independently reproduced.

EDUCATION

Asia Pacific University of Technology & Innovation (APU) — Kuala Lumpur, Malaysia
B.Sc. (Hons) Computer Science, dual award with De Montfort University, Leicester (UK) — Expected Nov 2026

- **GPA:** 3.7/4.0 | Vice-Chancellor's Award for Academic Excellence, 2023/2024

Lycée Français de Koweït — Kuwait City, Kuwait
French Baccalauréat, specialisation in Mathematics and Computer Science — 2023

- Graduated with highest honours, ranked 2nd of 80; Computer Science 19/20, Mathematics 18/20.

RESEARCH PROJECTS

AutoGuard AI: Acoustic Fault Detection and LLM Diagnostic System for Equitable Repair — 2026
Final Year Project, APU / De Montfort University | Supervisor: Ts. Dr. Manimegalai A/P Rajenderan

- Surveyed 30+ drivers to define the problem: 96.7% had felt overcharged by a mechanic and 76.7% found repair explanations confusing.
- Trained a CNN in TensorFlow/Keras on Mel-spectrograms to classify six fault types from audio, reaching 93% test accuracy on a public benchmark dataset.
- Deployed the model offline in a React Native app using TensorFlow Lite and INT8 quantisation (12x smaller), flagging uncertain readings instead of guessing.
- Built a RAG chatbot with Gemini, pgvector (Supabase PostgreSQL) and FastAPI that explains diagnoses in six languages, plus a fair-price estimator using World Bank PPP data.

RedSeaReefGap: Geospatial Sampling-Gap Analysis of Red Sea Reef-Fish Records — R

- Built an R pipeline that pulls GBIF Occurrence API records for six reef-fish families in the Red Sea and quantifies spatial and temporal sampling bias.
- Found that 70–85% of records per family come from a single country, skewed toward post-2015 citizen-science data.
- Made the analysis fully reproducible with independent validation checks and a GitHub Actions CI workflow.

AquaSize AI: Computer-Vision Pipeline for Reef-Fish Classification — Python

- Curated a multi-source reef-fish image dataset across 7 taxonomic families and trained a YOLO object-detection model in Python.
- Evaluated with confusion matrices and per-class recall on a held-out test split, exposing label inconsistencies in the source data.
- Deployed a live interactive classifier with fully documented methodology.

EXPERIENCE

Linux Lab Technician, Asia Pacific University — Jan 2026 – Present

- Lead hands-on Linux lab sessions for 20–30 students.
- Troubleshoot Linux networking, file sharing and system administration issues.

AI Software Engineer Intern, Keenify — Sep 2025 – Jan 2026

- Built a Christmas gift-box recommendation product end to end with Python/FastAPI and TypeScript, generating combinations from a 43-product catalogue with PDF report export; approved by the internship supervisor and two clients and launched on the client-facing website.
- Diagnosed a data-integrity bug in a production dashboard caused by mismatched source and stored data, and documented the root cause and fix.
- Built n8n automation workflows with webhooks, including a pipeline that turns Fathom meeting transcripts into shared-drive action-item documents with owners.
- Drafted a 50-page proposal for SCCCI on applying AI to its operational challenges.
- Delivered client-requested features and bug fixes in TypeScript: a print-ready calendar, job-posting management, mobile-responsive layouts and a database schema fix for employee offboarding.

TECHNICAL SKILLS

Programming: Python, R, SQL, TypeScript, JavaScript, Bash
Machine Learning & AI: TensorFlow/Keras, TensorFlow Lite, scikit-learn, CNNs, YOLO object detection, computer vision, INT8 quantisation, RAG, embeddings, prompt engineering, LLM APIs (Gemini, Groq)

Data & Signal Processing: NumPy, SciPy, librosa, Matplotlib, FFT, Mel-spectrograms, data augmentation, model evaluation (F1, recall, confusion matrices)
Research Methods: Survey design and quantitative analysis, sampling-bias analysis, geospatial analysis, literature review, reproducible pipelines, technical documentation
Tools & Infrastructure: Git/GitHub, GitHub Actions CI/CD, Linux, FastAPI, PostgreSQL, pgvector, Supabase, REST APIs (GBIF, World Bank), React Native/Expo, Render, n8n, webhooks, Figma

LEADERSHIP & ACTIVITIES

Professional Development Director (Board Member), Rotaract Club of APU Jun 2024 – Jun 2025

- Organised and delivered a hands-on web-development workshop for 60+ students, each building and publishing a portfolio website.

JomHack Hackathon

- Designed a multi-agent architecture to unify fragmented enterprise data silos, built under a no-AI-generated-code rule.

Student Ambassador, Asia Pacific University & ReSkills Apr 2025 – Jan 2026

- Represented the university at outreach events for prospective students and families.

CERTIFICATIONS & LANGUAGES

Certification: Red Hat System Administration I (RH124)
Languages: French (fluent), English (fluent), Spanish (intermediate)